Visual Profiling of KITTI Odometry Sequences Using Multi-Metric Difficulty Visualization

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Abstract

The KITTI odometry benchmark [Geiger et al., 2013] provides 11 real-world driving sequences that serve as a standard evaluation platform in SLAM research. A persistent limitation is that results are communicated almost exclusively as tables of scalar values, making it difficult to reason about multi-dimensional patterns, metric inter-relationships, and cluster structure across diverse driving environments. This paper presents a *visual profiling framework* that extracts seven difficulty metrics from ground-truth pose files to characterize *intrinsic sequence difficulty* independent of any specific algorithm. K-Means clustering (k=3, silhouette ≈ 0.61) identifies three tiers: Easy $= \{03, 04, 06\}$; Medium $= \{00, 02, 05, 07, 08, 09, 10\}$; Hard $= \{01\}$. Seq 01 achieves a composite score of 0.793—approximately $5.8\times$ the minimum—with 1,980 m accumulated drift. A Pearson correlation of r=0.95 between two metrics suggests potential redundancy in the standard set. PCA accounts for 94.3 % of variance in three interpretable components. These patterns are considerably more apparent through targeted visualization than from tabular data alone, supporting the role of visual analytics in benchmark analysis.

1 Introduction

Simultaneous Localization and Mapping (SLAM) estimates a robot's trajectory and map jointly from sensor data. Per-frame sensor noise compounds into *trajectory drift*; *loop closure*—recognizing a previously visited location—is the primary corrective mechanism. The KITTI benchmark [Geiger et al., 2012] collects centimetre-accurate GPS/IMU ground truth across 11 driving sequences spanning highway, urban, and campus environments (22,202 frames total), and has become the standard reference for odometry evaluation.

Despite widespread adoption, benchmark results are reported almost exclusively as tables of scalar errors. Questions such as "which sequences form natural difficulty groups?" or "do any metrics capture the same underlying property twice?" require reasoning across all $11 \times 7 = 77$ values simultaneously, a task that benefits substantially from visual support [Miller, 1956]. This work presents a framework for visual profiling of KITTI sequences based on ground-truth geometry alone.

Contributions.

1. Seven difficulty metrics extracted from raw pose files, independent of any estimated trajectory.
2. K-Means clustering (k=3, n_{init}=20) with silhouette-validated cluster count.
3. Five visualizations that surface patterns less apparent in tabular data (Table 2).
4. An open-source Python pipeline and interactive HTML dashboard with live metric weighting and animated trajectory replay.

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

2 Dataset and Methodology

Dataset. Table 1 summarises all 11 sequences. Seq 01 achieves the highest score (0.793); Seq 04 the lowest (0.137). Sequences 07 and 09 reach near-zero accumulated drift (9.5 m and 9.3 m) through near-complete loop closure.

Table 1: KITTI sequences with computed difficulty scores. **Bold**: Hard. *Italic*: minimum-score.

Seq	Frames	Score	Cluster	Step Var.	Drift (m)	Environment
00	4,541	0.258	Medium	0.271	97.2	Urban residential
01	1,101	**0.793**	**Hard**	**0.521**	**1,980**	Highway
02	4,661	0.444	Medium	0.233	924.5	Urban/campus
03	801	0.222	Easy	0.227	512.2	Suburban
04	271	*0.137*	*Easy*	*0.103*	393.6	Short highway
05	2,761	0.274	Medium	0.292	370.5	Urban residential
06	1,101	0.171	Easy	0.291	300.3	Urban mixed
07	1,101	0.355	Medium	0.321	9.5	Tight urban loop
08	4,071	0.276	Medium	0.286	312.0	Suburban mixed
09	1,591	0.271	Medium	0.261	9.3	Urban/campus
10	1,201	0.333	Medium	0.334	545.6	Campus/suburban

Feature Extraction. Seven metrics are derived from XZ ground-plane translations extracted from each pose file's 3×4 SE(3) matrix. (1) **Step-Distance Variability**: standard deviation of frame-to-frame distances, a motion-irregularity proxy—not equivalent to trajectory RMSE, which requires an estimated path. (2) **Drift Slope**: OLS regression coefficient of cumulative distance over frame index. (3) **Accumulated Drift**: terminal displacement $D = \|\mathbf{t}_N - \mathbf{t}_0\|$. (4–5) **Early/Late Error**: mean displacement in the first and second halves of the sequence. (6) **Path Curvature**: mean absolute curvature of the XZ trajectory via discrete differentiation; Seq 01 has $\bar{\kappa}$=0.005 (straight highway) vs. Seq 07 at $\bar{\kappa}$=23.49 (tight loop). (7) **Motion Entropy**: Shannon entropy of an 18-bin heading histogram, capturing directional diversity.

Normalisation and Composite Score. Each metric is Min-Max normalised to $[0, 1]$; the composite score is the equal-weighted mean $S_i = \frac{1}{7}\sum_j \hat{m}_{ij}$. Metric weights are adjustable in the interactive dashboard.

Clustering. K-Means with k=3, n_{init}=20, and k-means++ initialisation is applied to the 11×7 normalised matrix. The silhouette coefficient [Rousseeuw, 1987] peaks at ≈ 0.61 for k=3 over $k \in \{2, 3, 4, 5\}$, supporting the three-tier assignment.

3 Visualizations and Results

Five visualizations were designed to surface analytical patterns that are less readily apparent in tabular form.

Radar Profile (Fig. 1). The per-cluster radar chart displays all seven normalised metrics simultaneously. The Hard cluster (Seq 01) shows an asymmetric profile: high on all drift axes, near-zero on curvature. This shape conveys the physical mechanism of straight-line driving with no loop-closure opportunity more directly than any combination of scalar values.

Correlation Heatmap (Fig. 2). The 7×7 Pearson matrix reveals three patterns. Accumulated drift and late error share r=0.95 (R^2=0.90), suggesting substantial overlap; given N=11 this is exploratory, but raises a reasonable question about metric set redundancy. Drift slope and entropy show r=−0.62, consistent with the physical expectation that heading diversity limits monotonic drift accumulation. Step variability and curvature share r=0.12, supporting the retention of both.

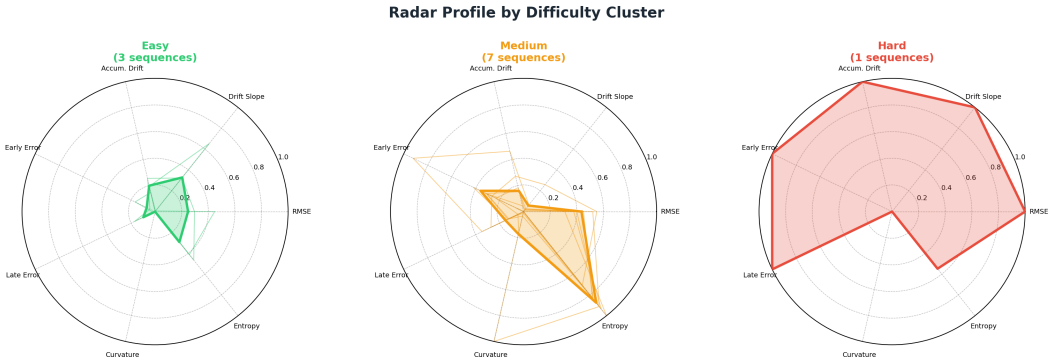

Figure 1: Per-cluster radar profiles. Easy (green): compact, low-valued polygon. Medium (orange): moderate profile with elevated entropy. Hard (red, Seq 01): asymmetric—high drift, near-zero curvature, encoding straight highway driving geometry.

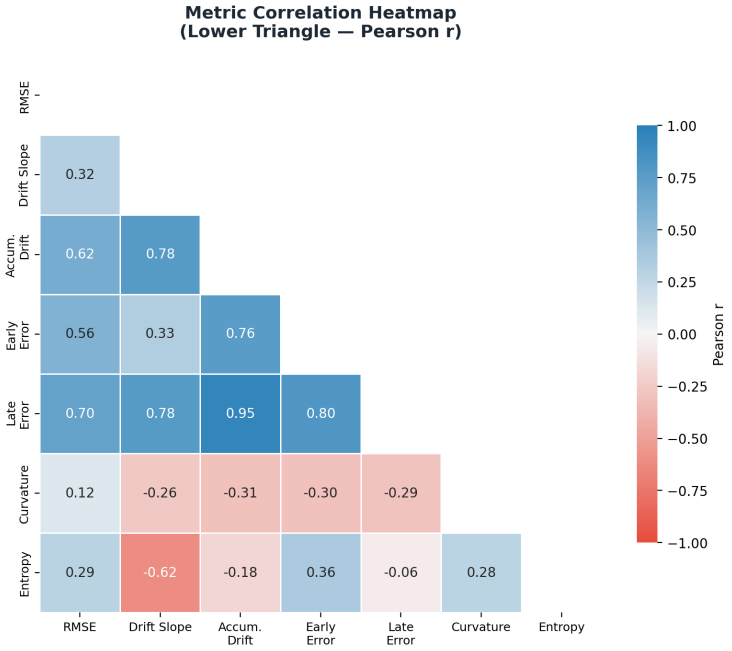

Figure 2: Pearson correlation heatmap (lower triangle). Blue: positive; red: negative. The r=0.95 cell (Accumulated Drift ↔ Late Error) is the highest off-diagonal value, suggesting potential redundancy in the metric set.

Difficulty Ranking (Fig. 3). The gap between Seq 01 (0.793) and Seq 02 (0.444) is 0.349, which exceeds the spread of the seven Medium sequences (0.186). This distributional feature, visible directly in the bar chart, suggests Seq 01 occupies a distinct region of the difficulty space rather than simply representing the extreme of a continuous distribution.

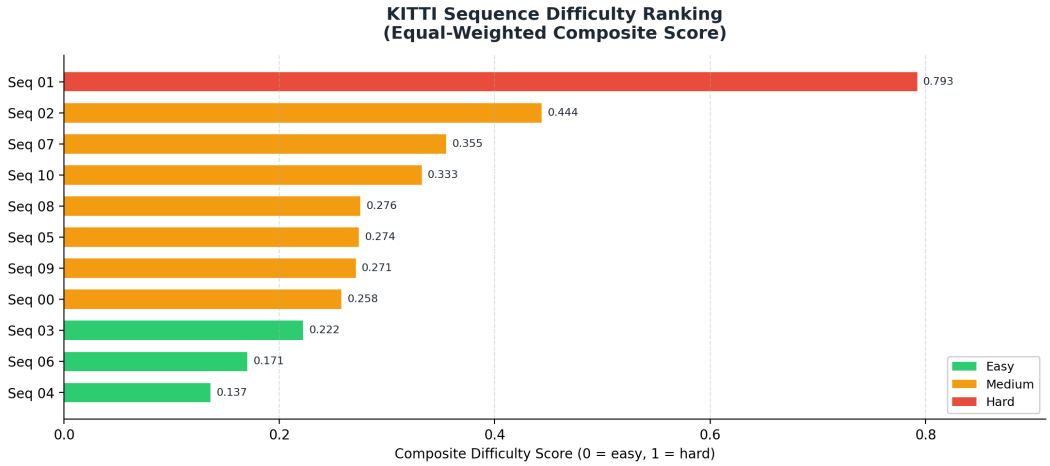

Figure 3: Composite difficulty ranking (equal-weighted). The 0.349 gap between Seq 01 and Seq 02 exceeds the full Medium cluster spread of 0.186.

3-D PCA Scatter (Fig. 4). Three PCA components account for 94.3 % of total variance: PC1 (53.9 %) loads on drift metrics; PC2 (26.6 %) on curvature and entropy; PC3 (13.8 %) on the temporal ratio. Seq 01 appears well-separated along PC1, and the three cluster groups are visually distinguishable in the projected space.

Runtime vs. Difficulty (Fig. 5). $r{=}0.042$ ($p{=}0.903$, $N{=}11$): no statistically significant correlation. With this sample size statistical power is limited, but the result is consistent with runtime being driven by frame count ($r{\approx}0.97$) rather than geometric complexity.

Temporal Error Growth (Fig. 6). Plotting displacement from origin over frame index reveals three qualitatively distinct patterns: (i) monotonic increase (Seq 01, 03, 10)—no self-correction; (ii) oscillating (Seq 00, 02, 05, 08, 09)—partial loop returns; (iii) V-shape return (Seq 06, 07)—near-complete loop closure. These patterns are not distinguishable from single-value drift summaries.

Ground-Truth Trajectories (Fig. 7). The XZ path plots provide direct geometric context: Seq 01's near-straight arc ($> 1{,}800$ m) illustrates why its drift metrics are extreme, while Seq 07's tight closed loop explains its near-zero accumulated drift.

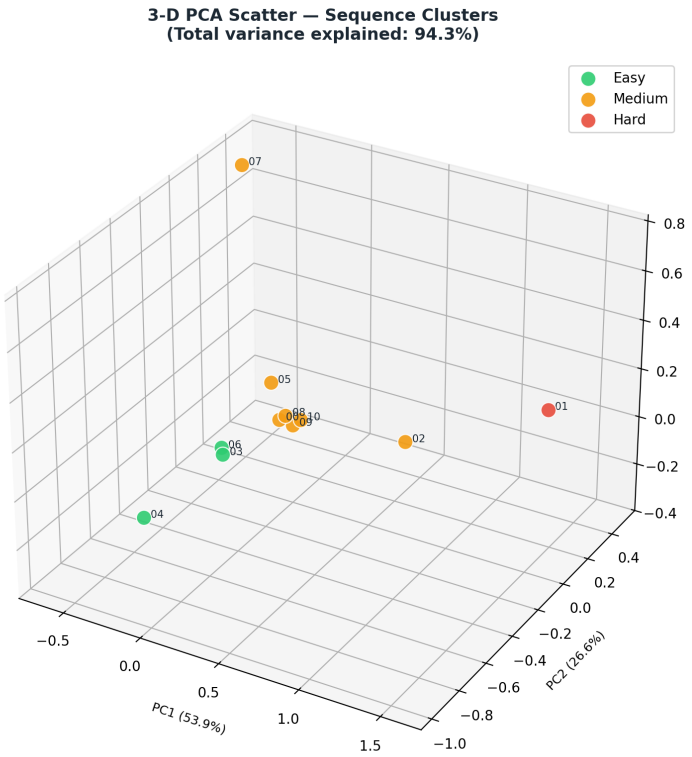

Figure 4: 3-D PCA scatter. Seq 01 (red) is separated along PC1 (drift, 53.9 %). Cumulative variance explained: 94.3 %.

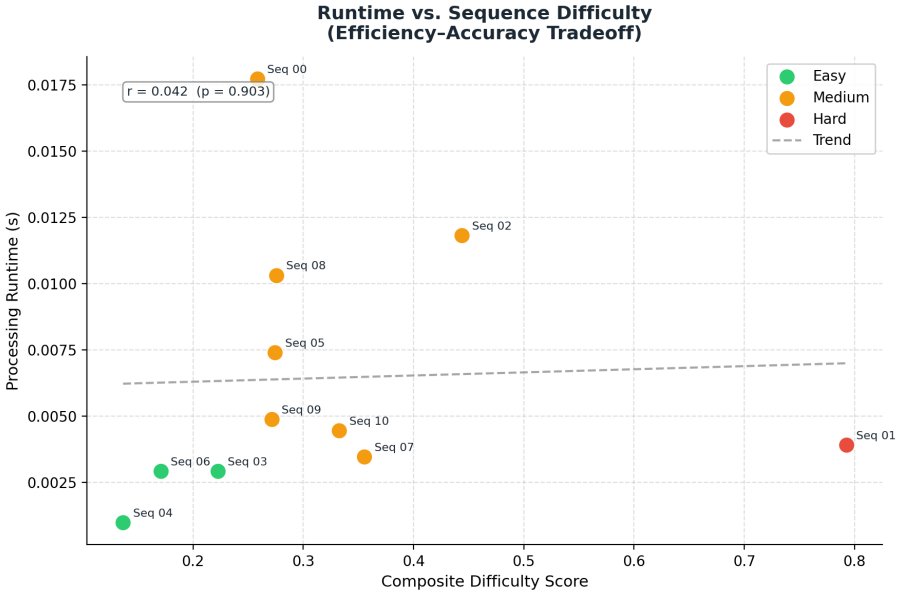

Figure 5: Runtime vs. difficulty (r=0.042, p=0.903). Frame count (r≈0.97) is a stronger predictor of runtime than difficulty.

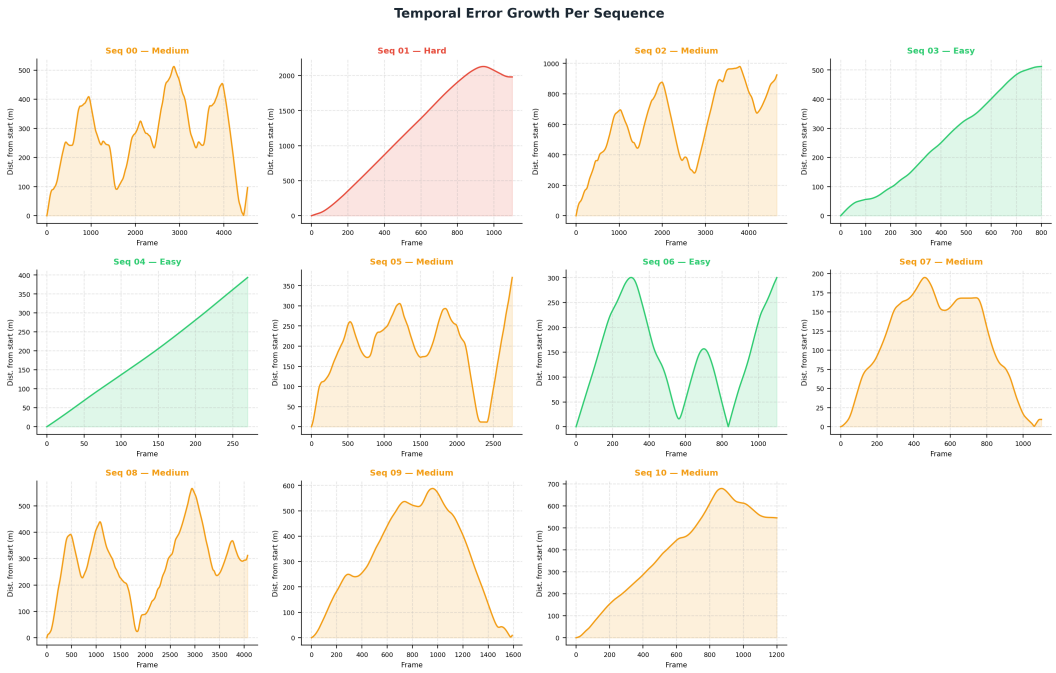

Figure 6: Distance from origin vs. frame index. Three patterns: monotonic (red); oscillating (orange); V-shape return (green). The same scalar drift summary does not distinguish these behaviours.

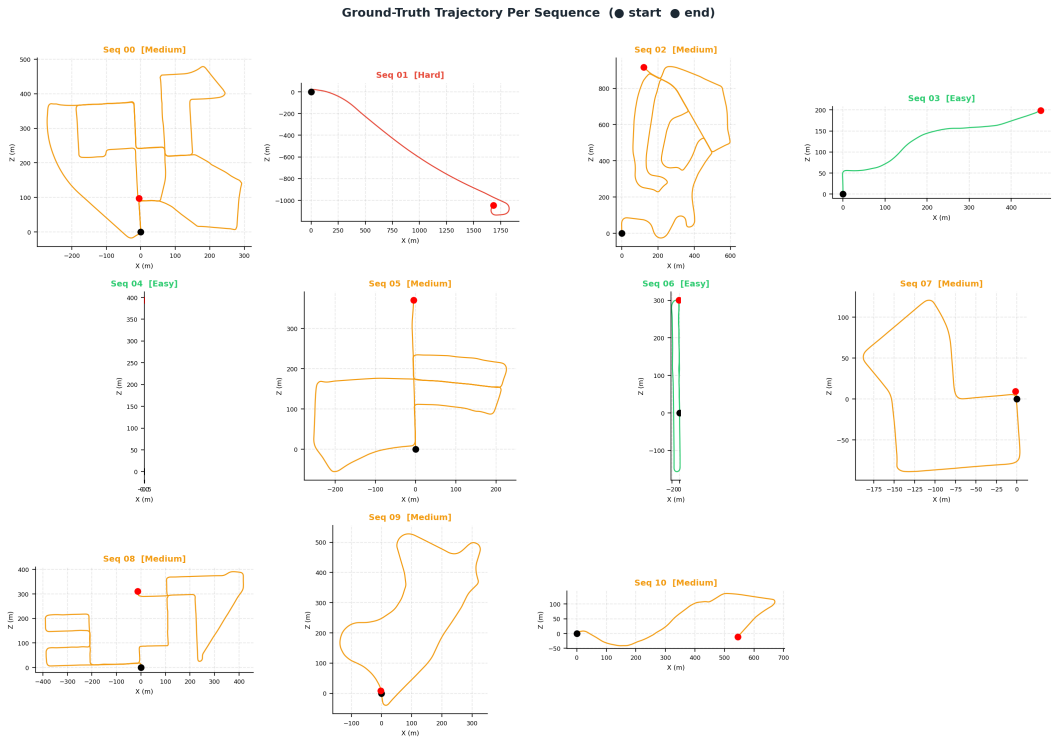

Figure 7: Ground-truth XZ trajectories. Black: start; red: end. Seq 01's straight arc and Seq 07's closed loop illustrate the geometric basis of their difficulty scores.

4 Visualization vs. Tabular Analysis

Table 2 compares six analytical questions against what tabular data and visualization each provide, following the logic of Anscombe's Quartet [Anscombe, 1973]. Four of the six questions involve information not readily available from the metrics table alone, illustrating the complementary role of visual analytics in benchmark interpretation.

Table 2: Analytical questions addressed by tabular data vs. the visualizations in Section 3.

Question	From Table	From Visualization
Which sequence is hardest, and by how much?	Sorting 11 values manually	Fig. 3: score magnitude and 0.349 gap visible directly
Are any metrics redundant?	Requires computing all pairs separately	Fig. 2: r=0.95 cell immediately apparent
Do sequences form natural groups?	Requires running a clustering procedure	Fig. 4: three groups visible in PCA projection
Is runtime related to difficulty?	Not in the metrics table	Fig. 5: flat trend, $r\approx0$
How does drift grow over time?	Not captured by any single metric	Fig. 6: three growth patterns
What does the driving path look like?	Not represented at all	Fig. 7: path geometry and loops visible

5 Discussion

Key Observations. Four findings emerge most clearly from the visualizations. (F1) Seq 01 has a composite score of 0.793 with a distributional gap (0.349) to the next sequence exceeding the full Medium cluster spread (0.186); its root cause—straight highway driving with $\bar{\kappa}$=0.005—is most clearly conveyed by the radar chart and trajectory plot. (F2) The r=0.95 correlation between accumulated drift and late error (R^2=0.90) is exploratory given N=11, but raises a question about whether both metrics need to be retained in future evaluation protocols. (F3) The absence of a significant runtime–difficulty relationship (r=0.042, p=0.903) suggests frame count is a more useful predictor of computational cost. (F4) Three-component PCA accounting for 94.3 % of variance with interpretable loadings indicates the metric set covers the main axes of variation in this dataset.

Limitations. With N=11, all statistical findings are exploratory and should not be generalised without further validation on larger benchmark collections. The framework measures intrinsic ground-truth geometry, not algorithm-specific difficulty, which depends additionally on factors such as feature availability and scene dynamics. Equal metric weighting is a simplifying assumption; the dashboard's weight interface supports sensitivity exploration.

6 Conclusion

This paper presented a visual profiling framework for the KITTI odometry benchmark applied to all 11 training sequences. The framework characterizes intrinsic sequence difficulty through seven geometry-based metrics without reference to any algorithm's output. The five visualizations surface patterns—distributional gaps, metric correlations, cluster structure, temporal growth dynamics, and path geometry—that are considerably less apparent from the raw metric table. As summarised in Table 2, visual analytics plays a complementary and practically useful role in benchmark analysis, supporting the interpretation of multi-dimensional data that summary statistics alone do not convey efficiently. The open-source pipeline and interactive HTML dashboard are directly applicable to any ground-truth pose dataset.

Acknowledgements. Completed for EECE 5642 Data Visualization, Prof. Y. Raymond Fu, Northeastern University.

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